

SC Honors Geometry - Nappier
Homework 10.1 – Lines and Segments That Intersect Circles

Name: _____

Vocabulary and basics.

1. A segment has both endpoints on a circle but does not pass through the center. What is it called? What if it DOES pass through the center?
2. A circle has radius 7. Find its diameter. Another circle has diameter 22. Find its radius.

Tangent or not? Justify with numbers.

3. A triangle has a radius of length 8 to point B, a segment $AB = 15$ from an outside point, and $AC = 17$ through the center region. Is AB tangent at B?
4. Same setup with radius 7, $AB = 10$, and $AC = 13$.

Tangent problems.

5. A hot-air balloon floats 2 mi above the Earth (radius ≈ 3960 mi). How far is it to the horizon along a tangent line? Round to the nearest tenth.
6. A tangent segment is 20 and the external piece of the line through the center is 10. Find the radius: $20^2 + r^2 = (r + 10)^2$.
7. Tangent segments from the same point measure $4x + 7$ and $6x - 9$. Find x and the length.
8. Tangent segments from the same point measure $2x + 3$ and $x + 9$. Find x and the length.
9. Two circles have the same center but different radii. What are they called, and how many common tangents do they have?
10. A point is 41 units from the center of a circle of radius 9. Find the length of a tangent segment from the point.